

ABOS Platform Specification v1.0

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The Official Technical Specification for the ABOS Aviation Intelligence Platform

Version: 1.0.0

Status: Draft

Type: Platform Specification

Normative: YES

Document Identifier: ABOS-PS-1.0

Normative Language

The key words **MUST**, **MUST NOT**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **MAY**, and **RECOMMENDED** are to be interpreted as described in RFC 2119.

Table of Contents

Part I — Foundation

1. Scope

Defines the normative architecture of the ABOS Platform.

Specifies:

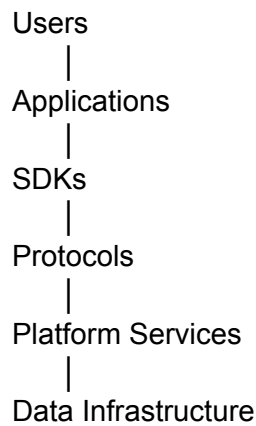
- Platform Components
- Service Boundaries
- Protocol Relationships
- Data Model
- AI Architecture
- Security

- Deployment
- Governance
- Certification

This specification is the authoritative source for all ABOS implementations.

2. Platform Overview

Defines the complete logical architecture.



Every compliant implementation SHALL conform to this layered architecture.

3. Core Principles

Normative versions of Vision principles.

Examples:

PS-3.1

Every platform capability SHALL be exposed through a documented interface.

PS-3.2

Business logic MUST NOT exist exclusively inside presentation layers.

PS-3.3

Applications SHALL consume Platform Services.

Applications SHALL NOT define Platform Services.

PS-3.4

All public interfaces SHALL be versioned.

PS-3.5

Every externally accessible service SHALL expose machine-readable contracts.

Part II — Platform Architecture

4. Reference Architecture

Defines every platform layer.

Presentation Layer

↓

Application Layer

↓

SDK Layer

↓

Protocol Layer

↓

Platform Layer



Infrastructure Layer

5. Platform Components

Each component SHALL define:

Purpose

Responsibilities

Interfaces

Dependencies

Lifecycle

Version

Owner

Conformance Level

Platform Components include:

Core API

APL

Knowledge Graph

Reasoning Engine

Event Bus

SDK Layer

MCP Integration

Identity Service

Marketplace Service

Developer Portal

Certification Service

6. Platform Dependency Rules

Normative rules.

Example:

Applications SHALL NOT directly access platform databases.

Applications MUST use Platform APIs.

Protocols SHALL remain implementation-independent.

Platform Services SHALL remain vendor-neutral.

Part III — Platform Services

7. Identity Service

Normative specification.

Defines:

Users

Organizations

Developers

Agents

API Keys

OAuth

OIDC

Namespaces

Certificates

8. Aircraft Service

Defines Aircraft entity.

Aircraft Identity.

Aircraft Lifecycle.

Aircraft Relationships.

9. Listing Service

Defines Listing lifecycle.

State transitions.

Ownership.

Publication.

Discovery.

10. Search Service

Defines Intent Search.

Natural Language Search.

Structured Search.

Ranking.

Semantic Search.

11. Intelligence Service

Defines:

ATI

Valuation

Market Intelligence

Buyer Matching

Recommendations

Confidence

Evidence

12. Messaging Service

Conversation model.

Notifications.

Events.

Communication.

13. Integration Service

External APIs.

Import.

Export.

Synchronization.

Partner Integration.

Part IV — Protocol Specifications

Platform SHALL include these protocol families.

14. APL

Reference:

APL Specification

15. OpenAPI

Reference:

ABOS OpenAPI Specification

16. Event Protocol

Reference:

ABOS Event Specification

17. Knowledge Graph

Reference:

Knowledge Graph Specification

18. ATI

Reference:

ATI Specification

19. MCP Integration

Reference:

MCP Integration Specification

Part V — AI Platform

20. AI Runtime

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Agent Runtime

Reasoning

Planning

Tool Invocation

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Agent Identity

Agent Lifecycle

Agent Registry

Capabilities

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22. Reasoning Engine

Intent Resolution

Planning

Evidence

Confidence

Decision Graph

23. Marketplace

Discovery

Publication

Trust

Certification

Revenue

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24. Identity

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Developer Identity

Agent Identity

Organization Identity

25. Authorization

RBAC

ABAC

Policies

Scopes

Delegation

26. Trust

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Audit

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27. Compliance

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Revocation

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.NET

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35. Observability

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Logging

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AI Decisions

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36. Versioning

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Protocol Version

API Version

SDK Version

APL Version

RFC Version

37. Compatibility

Backward Compatibility

Forward Compatibility

Migration

Deprecation

38. Conformance

Core Platform

Core API

APL

SDK

AI Runtime

Marketplace

MCP

Each implementation SHALL declare a Conformance Profile.

39. Certification

SDK Certification

Partner Certification

Agent Certification

Enterprise Certification

Reference Implementation Certification

Part X — Ecosystem

40. Applications

Reference applications include:

- Marketplace
- Mobile
- ChatGPT App
- Facebook App
- Dealer CRM
- OEM Portal
- Banking
- Insurance
- MRO

These applications are **not** part of the platform core.

They are reference consumers of the platform.

41. Developer Portal

Documentation

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API Explorer

Certification

42. Roadmap

Future Platform Versions

Protocol Evolution

Experimental Specifications

Future RFC Series

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- Appendix A — OpenAPI Reference
- Appendix B — JSON Schemas
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Strategický návrh

Myslím, že **ABOS Platform Specification v1.0** by neměl být pouze jeden dlouhý dokument. Stejně jako Kubernetes nebo OpenTelemetry bych z něj udělal **hlavní vstupní dokument**, který odkazuje na samostatné normativní specifikace (APL, API, ATI, Event, MCP, Security atd.). To umožní jednotlivé části verzovat nezávisle, aniž by bylo nutné vydávat novou verzi celé platformy při každé změně jednoho protokolu. Taková struktura bude lépe škálovat s růstem ABOS jako otevřené platformy.

Appendix A

Podle mě by právě **Appendices** měly být nejcennější částí celé platformy. Neměly by být jen přílohou, ale **normativní referenční knihovnou ABOS**.

Navrhl bych je rozšířit takto.

Appendix A — OpenAPI Reference

Status: Normative

Purpose

Defines the complete REST API contract for the ABOS Platform.

Contents

- API Overview
- Authentication
- Versioning
- Endpoints
- Operations
- Request Schemas
- Response Schemas
- Pagination
- Filtering
- Sorting
- Error Responses
- Examples
- Webhooks

- Streaming APIs

Artifacts:

openapi.yaml

openapi.json

Postman Collection

Appendix B

Appendix B — JSON Schemas

Status: Normative

Purpose

Defines every canonical JSON object used by the platform.

Core Schemas

Aircraft

AircraftIdentity

AircraftListing

Organization

User

Agent

Capability

Intent

Mission

Evidence

Confidence

ATI

Valuation

MarketSignal

Event

Notification

Webhook

Error

Each schema includes:

- JSON Schema
- Required fields
- Validation rules
- Examples
- Version history

Appendix C

Appendix C — APL Schemas

Status: Normative

Defines all APL protocol objects.

Includes

Agent Manifest

Capability Manifest

Permission Manifest

Audit Manifest

Identity Manifest

Trust Manifest

Certificate

Registry Entry

Communication Envelope

Invocation Request

Invocation Response

Also:

APL URI Specification

APL Namespace Rules

APL Version Format

Appendix D

Appendix D — Event Catalog

Status: Normative

Defines every platform event.

Grouped by domain.

Aircraft Events

AircraftCreated

AircraftUpdated

AircraftArchived

Listing Events

ListingCreated

ListingPublished

ListingUpdated

ListingExpired

ListingSold

Intelligence Events

ATIComputed

ValuationCompleted

BuyerMatched

MarketSignalGenerated

Security Events

PermissionDenied

CertificateRevoked

AuthenticationFailed

TrustViolation

Agent Events

AgentRegistered

AgentInvoked

AgentCompleted

AgentFailed

AgentCertified

Each event specifies:

- Event ID
 - Payload
 - Producer
 - Consumer
 - Version
 - Delivery guarantees
-

Appendix E

Appendix E — Agent Catalog

Status: Informative + Normative Registry

Defines official ABOS agents.

Example:

Aircraft Discovery Agent

Aircraft Valuation Agent

Buyer Matching Agent

Market Intelligence Agent

Compliance Agent

Maintenance Intelligence Agent

Insurance Risk Agent

Finance Advisor

Each entry includes:

- Agent ID
- Version
- Owner

- Skills
 - Required Permissions
 - Trust Level
 - Supported Models
 - MCP Compatibility
 - SDK Support
-

.

Appendix F

Appendix F — MCP Tools

Status: Normative

Defines official MCP tools exposed by ABOS.

Examples:

search_aircraft

valueate_aircraft

calculate_ati

find_buyers

market_analysis

compare_aircraft

estimate_operating_cost

generate_listing

verify_aircraft_identity

For each tool:

- Tool Name

- Description
 - Input Schema
 - Output Schema
 - Required Permissions
 - Example Invocation
 - Streaming Support
-

Appendix G

Appendix G — Error Catalog

Status: Normative

Every platform error receives a permanent identifier.

Example:

ABOS-1000

Authentication Failed

ABOS-1100

Permission Denied

ABOS-2100

Aircraft Not Found

ABOS-3100

Invalid Capability

ABOS-4100

APL Validation Error

Each error defines:

- Code
- HTTP Status
- Description

- Cause
 - Recovery
 - Retry Policy
-

Appendix H

Appendix H — Glossary

Status: Normative

Canonical definitions.

Examples:

Aircraft

Aircraft Identity

Digital Twin

ATI

Knowledge Graph

Mission

Intent

Capability

Evidence

Confidence

Agent

Skill

Reasoning

Marketplace

APL

MCP

SDK

Platform Service

Reference Implementation

Every specification references this glossary.

Appendix I

Appendix I — Reference Implementations

Status: Informative

Defines official implementations maintained by the ABOS project.

Includes

Reference API Server

Reference SDK

Reference Registry

Reference MCP Server

Reference Marketplace

Reference Mobile App

Reference CLI

Reference Event Bus

Reference Knowledge Graph

Each implementation documents:

- Repository
 - Version
 - Supported Specifications
 - Conformance Profile
 - Deployment Guide
-

Appendix J

Appendix J — Security Guidelines

Status: Normative

Defines platform-wide security requirements.

Topics

Identity

- OAuth 2.1
- OpenID Connect
- API Keys
- Mutual TLS

Authorization

- RBAC
- ABAC
- Capability Policies

Trust

- Digital Signatures
- Certificates
- Trust Chain

AI Security

- Prompt Injection Protection
- Model Isolation

- Tool Authorization
- Data Provenance
- Explainability

Infrastructure

- Secrets Management
- Encryption at Rest
- Encryption in Transit
- Key Rotation
- Audit Logging

Incident Response

- Revocation
 - Security Events
 - Recovery Procedures
-

Appendix K

Appendix K — Conformance Test Suite

- API Conformance Tests
- APL Conformance Tests
- MCP Conformance Tests
- SDK Conformance Tests
- Event Conformance Tests
- Security Conformance Tests
- AI Agent Conformance Tests
- Certification Test Profiles

Každá implementace (např. Reference API, JavaScript SDK, Python SDK nebo externí partner) by musela projít touto sadou testů, aby mohla deklarovat kompatibilitu s **ABOS Platform Specification**.